

Davide Capone

Date of birth: 23/04/1999

Nationality: Italian

CONTACT INFORMATION

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PROFILE DESCRIPTION

I am a Data Science and Scientific Computing student with a growing foundation in artificial intelligence, data analysis, and scientific problem-solving. My academic journey began with a bachelor's degree in *Internet of Things, Big Data & Web*, where I gained valuable insights into modern technologies and their applications. I am currently pursuing a master's degree in *Data Science and Scientific Computing*, focusing on advanced data analysis, artificial intelligence, and computational methods. I have also had the opportunity to develop my communication and presentation skills as a *Computer Science teacher* at a secondary school, enabling me to explain technical concepts effectively to diverse audiences. I am motivated by a genuine passion for computer science and mathematics, and I am eager to contribute to innovative AI projects, continue learning, and grow as a professional in the field.

EXPERIENCE

Deep Learning Engineer (Intern)

Danieli Research Center, Buttrio (Udine), ITA

2024/12 → Present

What I am doing here?

- Building and optimizing existing deep learning models by leveraging state-of-the-art algorithms to improve performance and accuracy.
- Studying and utilizing machine learning frameworks tailored for computer vision tasks, ensuring efficient implementation and scalability.
- Researching and implementing advanced object tracking algorithms to address specific industrial challenges.
- Collaborating with team members to integrate optimized models into operational systems and evaluate their performance in production environments.

Computer Science Teacher

ISIS Paschini Linussio, Raimondo D'Aronco, (Udine), ITA

2023/10 → 2024/09

- Taught computer science, programming, and ICT technologies to secondary school students.
- Designed lesson plans combining theory and practice to ensure a comprehensive understanding of core topics.
- Fostered collaboration and critical thinking through group projects, coding challenges, and real-world problem-solving.

Computer Vision Engineer (Intern)

Danieli Research Center, Buttrio (Udine), ITA

2022/10 → 2022/12

- Computer vision techniques, including calibration methods such as homography estimation for camera alignment, object detection and object localization, with a focus on addressing real-world industrial challenges.
- Researched and implemented state-of-the-art algorithms to optimize detection and localization of objects in a steel slab storage yard.
- Created datasets using motion detection algorithms and YOLO for bounding box management, tailored to industrial-scale requirements.
- Designed, implemented, and tested convolutional neural networks (CNNs) for object localization and detection.

- The analysis of model performances with hyperparameter tuning methods was undertaken, with the objective of identifying and deploying the most effective solutions in a production setting.

EDUCATION

Master's Degree in Data Science and Scientific Computing

University of Trieste, (Trieste), ITA

2023/10 → Present

The course is jointly offered by the *University of Trieste, University of Udine, ICTP* and *SISSA*..

- Key courses: *Advanced Programming and Algorithmic Design, Data Management for Big Data, High Performance Computing, Numerical Analysis, Probabilistic Machine Learning, Reinforcement Learning, Statistical Methods for Data Science, Deep Learning, Information Retrieval and Data Visualization.*

Bachelor's Degree in Internet of Things, Big Data & Web

University of Udine, (Udine), ITA

2019/10 → 2022/12

Final Grade: 100/110 — Thesis: *Applications of Computer Vision for Object Detection and Localization in a Steel Yard Plant.*

- Key courses: *Algorithms and Data Structures, Machine Learning, Data Science, Applied Statistics, Social Computing, Mathematical Analysis, Social Computing, Object Oriented Programming, Cloud Computing, Software Engineering, Databases.*